

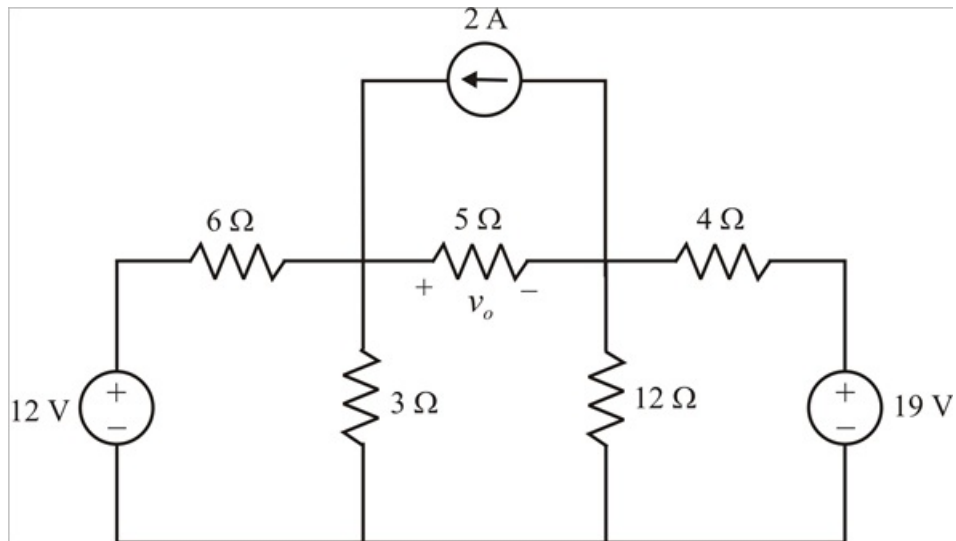
Problem

Use Thevenin's theorem to find v_o in Prob. 4.12.

Step-by-step solution

Step 1 of 11

Given circuit:

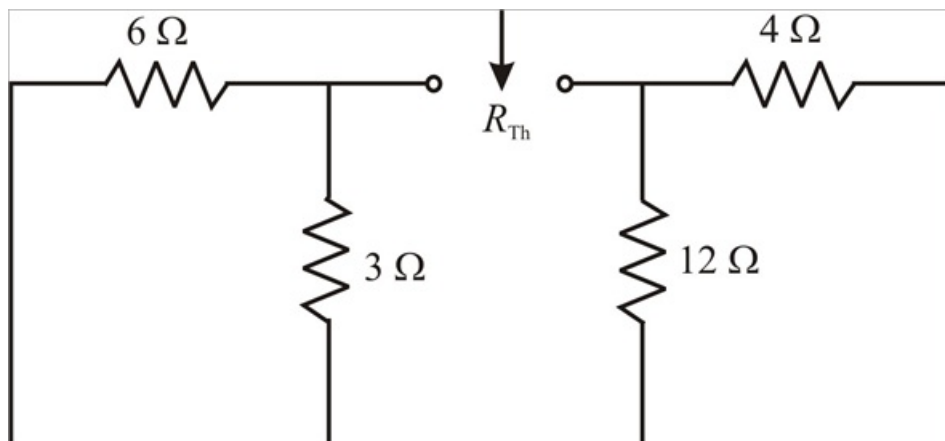


Step 2 of 11

We need to find voltage v_o using Thevenin's theorem. For that we need to find Thevenin resistance R_{Th} and Thevenin voltage V_{Th} across 5Ω resistor.

Step 3 of 11

To get R_{Th} , replace voltage sources by short circuit and current source by open circuit. Then the above figure will be modified as shown below



Step 4 of 11

In the above circuit, resistances 6Ω and 3Ω are in parallel. Hence its equivalent resistance is given by

$$\Rightarrow 6 \parallel 3$$

$$\Rightarrow \frac{(6)(3)}{6+3}$$

$$\Rightarrow \frac{18}{9}$$

$$\Rightarrow 2\Omega$$

Similarly resistances 12Ω and 4Ω are in parallel. Hence its equivalent resistance is

$$\Rightarrow 12 \parallel 4$$

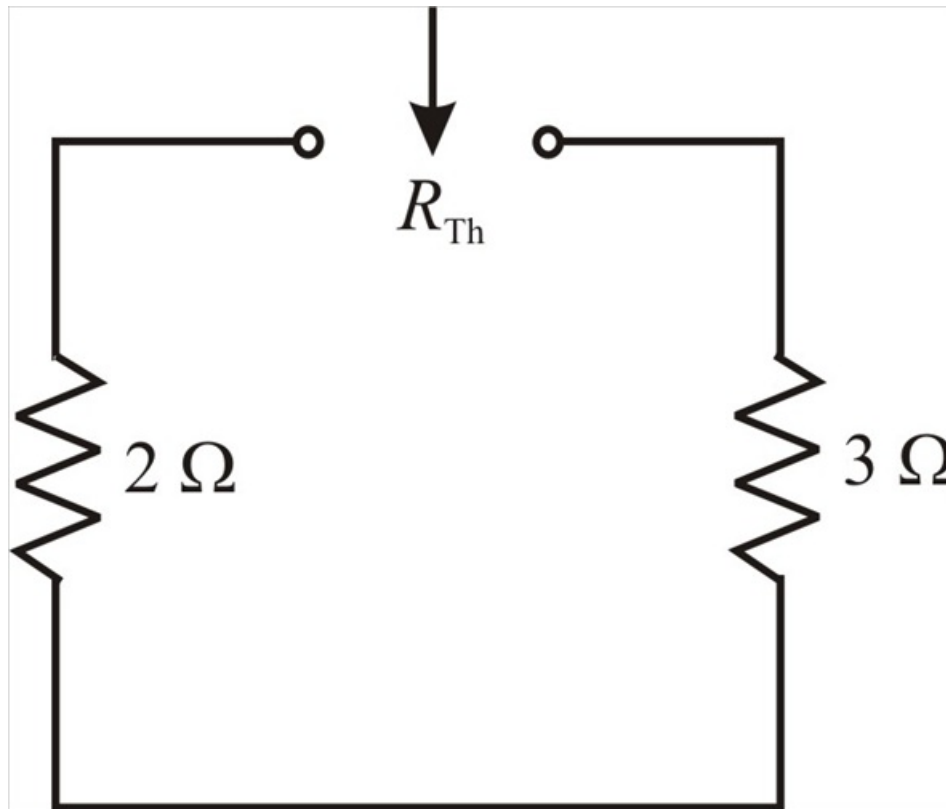
$$\Rightarrow \frac{(12)(4)}{12+4}$$

$$\Rightarrow \frac{48}{16}$$

$$\Rightarrow 3\Omega$$

Step 5 of 11

Now the above circuit is modified as shown below



Step 6 of 11

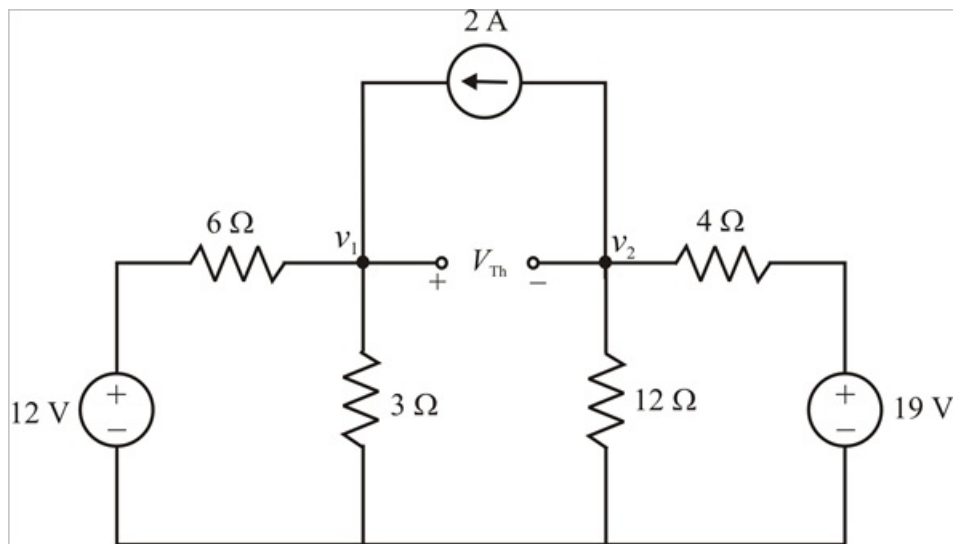
In the above circuit, resistances 2Ω and 3Ω are in series. Hence equivalent thevenin resistance is given by

$$R_{Th} = 2 + 3$$

$$R_{Th} = 5\Omega$$

Step 7 of 11

To get V_{Th} , consider the following circuit



By applying Kirchhoff's current law at node 1, we get

$$-2 + \frac{v_1 - 12}{6} + \frac{v_1}{3} = 0$$

$$\frac{v_1 - 12}{6} + \frac{v_1}{3} = 2$$

$$\frac{v_1 - 12 + 2v_1}{6} = 2$$

$$3v_1 - 12 = 12$$

$$v_1 = \frac{24}{3}$$

$$v_1 = 8 \text{ V}$$

Step 8 of 11

By applying Kirchhoff's current law at node 2, we get

$$2 + \frac{v_2}{12} + \frac{v_2 - 19}{4} = 0$$

$$\frac{v_2}{12} + \frac{v_2 - 19}{4} = -2$$

$$\frac{v_2 + 3(v_2 - 19)}{12} = -2$$

$$4v_2 - 57 = -24$$

$$4v_2 = 33$$

$$v_2 = 8.25 \text{ V}$$

Step 9 of 11

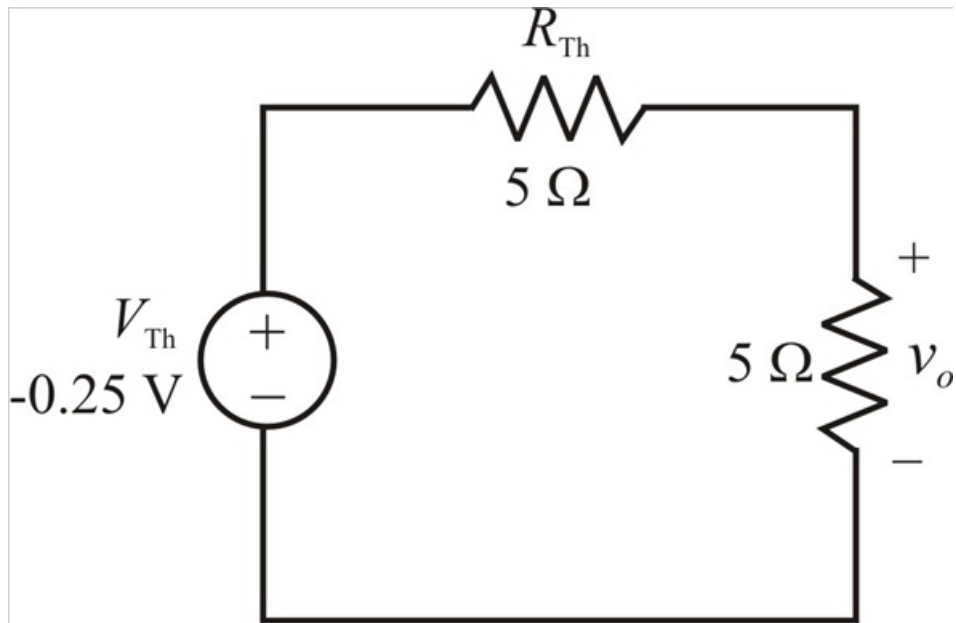
Now Thevenin equivalent voltage is given by

$$\begin{aligned} V_{Th} &= v_1 - v_2 \\ &= 8 - 8.25 \end{aligned}$$

$$V_{Th} = -0.25 \text{ V}$$

Step 10 of 11

Thevenin's equivalent network across 5Ω resistor can be represented by the following circuit.



Step 11 of 11

By applying voltage division to the above circuit, we get

$$\begin{aligned} v_o &= \left(\frac{5}{5+5} \right) (V_{Th}) \\ &= \left(\frac{5}{5+5} \right) (-0.25) \\ &= \left(\frac{1}{2} \right) (-0.25) \\ &= -0.125\text{ V} \\ v_o &= -125\text{ mV} \end{aligned}$$

Therefore, the voltage v_o in the given circuit is $\boxed{v_o = -125\text{ mV}}$.